



OPEN Experiments and physical Investigation on the adsorption of methylene on activated carbon

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In this study, activated carbon (AC) derived from olive waste was employed as an alternative adsorbent for the removal of methylene blue (MB) from aqueous solutions. The AC was characterized using FTIR, SEM, XRD and N₂ adsorption–desorption analyses. The pseudo-second-order kinetic model and the Langmuir adsorption isotherm provided a good fit to the experimental data at 298, 303, and 313 K. A multilayer adsorption model was applied to elucidate the adsorption mechanisms between the dye molecules and the adsorbent. The number of MB active sites (*n*) was greater than unity at all temperatures, indicating a vertical orientation of the adsorbed molecules. The number of active sites (*D_m*) showed only slight variation with temperature. The decrease in MB adsorption capacities (*Q_m*) from 235.21 to 198.62 mg·g⁻¹ over the temperature range of 298–313 K was primarily influenced by the *n* parameter. Adsorption energy analysis revealed that MB adsorption was exothermic and occurred via a physisorption process. This study provides valuable insights into the interactions between MB molecules and AC active sites.

Keywords Adsorption, Activated carbon, Methylene blue, Physical modeling

Many industries generate effluents containing dyes during their manufacturing processes¹. In particular, textile industries discharge significant amounts of dyes in their wastewater, which exhibit toxic effects on the environment and living organisms². Various chemical and physical methods have been developed for dye removal, including coagulation, adsorption, membrane separation, and chemical oxidation³. Among these treatment techniques, adsorption is recognized as one of the most effective methods⁴. Numerous adsorbents have been tested for color removal from aqueous effluents, with activated carbon being the most widely used due to its excellent textural properties; however, its high production and regeneration costs limit its large-scale application⁵. Consequently, many researchers have explored the use of low-cost materials as carbon precursors for dye and wastewater treatment, such as corn cob⁶, *Coriandrum sativum*⁷, waste orange peels⁸, almond shells⁹, peach stones¹⁰, and pine wood¹¹.

In this context, activated carbon derived from olive waste was used to study the removal of methylene blue (MB) dye. Adsorption isotherms for MB on the activated carbon were measured at different temperatures (298, 308, and 318 K), and the adsorbent was thoroughly characterized. The main objective of this study was to investigate the adsorption mechanism through both experimental and theoretical analyses. A physical model was successfully applied to establish a strong correlation between the adsorption properties, the adsorbent characterization, and the MB adsorption behavior. These findings provide new insights into the separation mechanism and contribute to a better understanding of the methylene blue adsorption process.

Experimental section Synthesis of activated carbon

Activated carbon (AC) was prepared from olive waste following a modified procedure¹². The raw olives were washed with demineralized water and dried in an oven at 100 °C for 24 h. The dried material was then ground

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to particles of 200–400 μm . The olive particles were impregnated with 85% phosphoric acid (H_3PO_4) at a 1:1 weight ratio for 24 h. The impregnated material was heated at 100 $^\circ\text{C}$ to remove excess moisture, then washed with deionized water and NaHCO_3 to neutralize residual acidity. Carbonization was performed at 600 $^\circ\text{C}$ for 4 h, and the resulting AC was washed thoroughly with distilled water and dried at 100 $^\circ\text{C}$ for subsequent use.

Characterization of the adsorbent

Various techniques were employed to characterize the prepared activated carbon. The crystallinity of the adsorbent was analyzed by X-ray diffraction (XRD) using a Philips MPD1880-PW1710 diffractometer. The functional groups were identified by Fourier transform infrared (FTIR) spectroscopy (Bruker Vertex 70-RAM II). The surface morphology of the activated carbon was examined using a scanning electron microscope (SEM, FEI Q250 Thermo Fisher). The textural properties were determined by nitrogen adsorption–desorption analysis using the Brunauer–Emmett–Teller (BET) method with a Micromeritics 3Flex analyzer.

Study of adsorption

Adsorption kinetics

The adsorption experiments were conducted as follows: 50 mL of the model solution, containing 100 mg L^{-1} of methylene blue (MB), was placed in a glass vessel, and the pH was adjusted using dilute HNO_3 . The pH values studied ranged from 2 to 10. A predetermined amount of adsorbent (0.01–0.9 g) was then added to the solution, followed by sonication at room temperature for 5 to 180 min. The absorbance of the solutions was measured using a UV spectrophotometer at $\lambda_{\text{max}} = 664 \text{ nm}$, corresponding to the maximum wavelength of MB.

The equilibrium adsorption capacity q_e is calculated using the following Eq. (1)¹²:

$$q_e = \frac{(C_0 - C_e) V}{w} \quad (1)$$

where C_0 , C_e , V and w are the initial methylene blue concentration (mg L^{-1}), the final methylene blue concentration (mg L^{-1}), the solution volume (l) and the mass of AC (g), respectively.

Kinetic models

To examine the rate of the adsorption process, the pseudo-first order kinetic models and the pseudo-second order kinetic model are given by the following Eq. 4 and 5, respectively¹²:

$$\ln(q_e - q_t) = \ln q_e + k_1 t \quad (2)$$

$$\frac{t}{q_t} = \frac{1}{k_2 q_e^2} + \frac{t}{q_e} \quad (3)$$

where $q_e (\text{mg g}^{-1})$ was the adsorption capacity in equilibrium. $q_t (\text{mg g}^{-1})$ are adsorption capacity in any time (t , min). $k_1 (\text{min}^{-1})$ and $k_2 (\text{g mg}^{-1} \text{min}^{-1})$ was the equivalent adsorption velocity constants for pseudo-first-order and pseudo-second-order respectively.

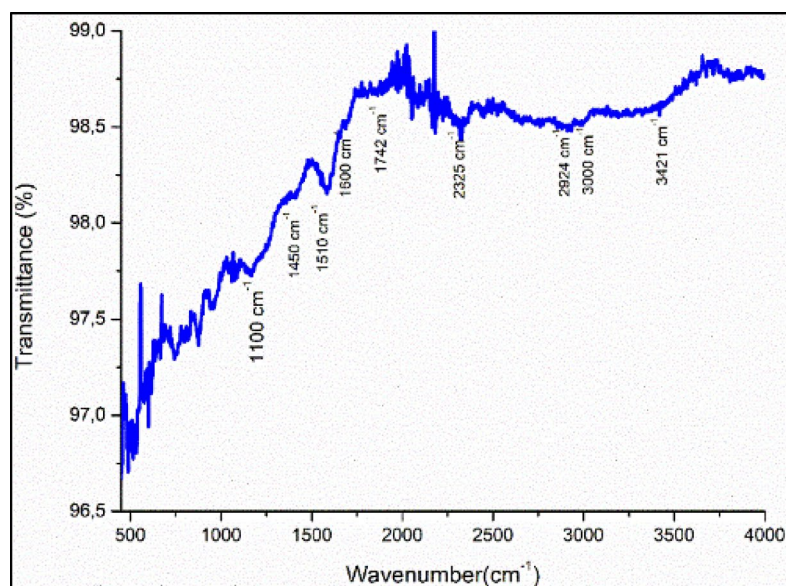


Fig. 1. FTIR spectra of the AC^{5,26}.

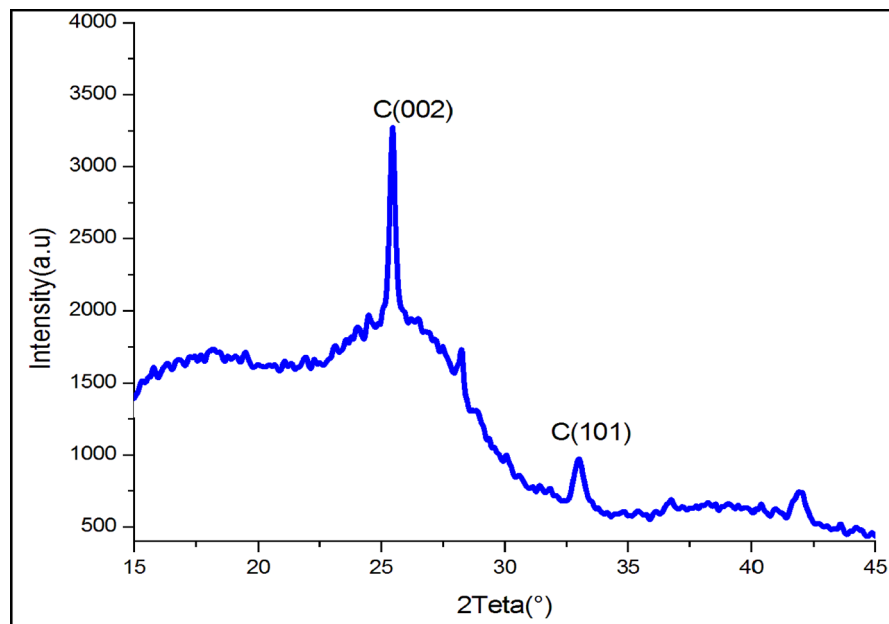


Fig. 2. XRD of the AC^{5,26,28}.

Isotherm modeling

Classical models

The classical isotherm model used to fit the adsorption isotherm was the Langmuir (Eq. 4) and Freundlich models (Eq. 4)¹²:

$$q_e = q_{max} K_L \frac{C_e}{1 + K_L C_e} \quad (4)$$

$$q_e = K_F C_e^{1/n} \quad (5)$$

C_e , K_L and K_F represent respectively the concentration at equilibrium in (mg L^{-1}), Langmuir and Freundlich constant.

To further elucidate the mechanisms governing MB adsorption on Cs, advanced statistical physics models were employed. These models were used to complement and validate the results of classical adsorption models, providing a deeper understanding of the adsorbent-adsorbate interaction dynamics¹³.

A multilayer physical model was employed to characterize the adsorption systems (Eq. 6). This model assumes that the adsorption of MB onto AC occurs as a multilayer process, involving the formation of a limited number of adsorbate layers. In characterizing the adsorption interactions, the model provides two distinct adsorption energies: one corresponding to dye-adsorbent surface interactions and another related to dye-dye interactions. The first adsorption energy is associated with the formation of the initial adsorbate layer, while the second arises from the formation of additional, yet limited, adsorbate layers. According to this model, the total number of dye molecule layers formed is expressed as $1 + N_2$ ¹⁴.

The expression of this model is defined as a function of the receptor site density (D_m), the number of MB molecules attached per site (n), and two half-saturation concentrations, C_1 and C_2 , which correspond to the formation of the first and second adsorbate layers, respectively. It is expressed by the following equation¹⁴:

$$Q_e = n D_m \frac{\left(\frac{C_e}{C_1}\right)^n + 2\left(\frac{C_e}{C_1}\right)^{2n}}{1 + \left(\frac{C_e}{C_1}\right)^n + \left(\frac{C_e}{C_2}\right)^{2n}} \quad (6)$$

Both half-saturation concentrations are model parameters directly related to the adsorption energies. The corresponding expressions of the adsorption energies are given by¹⁴:

$$\Delta E_1 = RT \ln \left(\frac{C_s}{C_1} \right) \quad (7)$$

$$\Delta E_2 = RT \ln \left(\frac{C_s}{C_2} \right) \quad (8)$$

where C_s (mg L^{-1}) was the water solubility of MB dye.

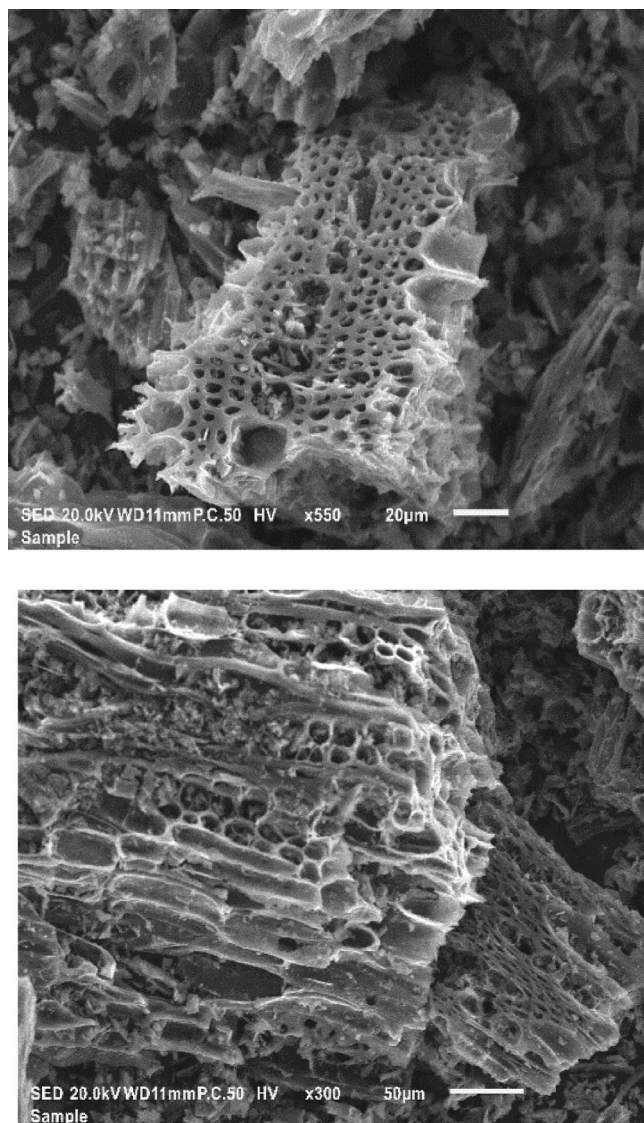


Fig. 3. SEM of the AC^{5,26,27}.

S_{BET} (m ² /g)	1565
V_{mic} (cm ³ /g)	0.649
V_{mes} (cm ³ /g)	0.027
V_{total} (cm ³ /g)	0.676

Table 1. Textural parameters of activated carbon^{5,26}.

Results and discussion

Characterization of the AC

The FTIR spectra of the activated carbon are shown in Fig. 1. The band at 3420 cm⁻¹ was attributed to –OH vibrations. The bands at 2924 cm⁻¹ and 3000 cm⁻¹ correspond to –CH₂ and aliphatic –CH groups, respectively. Additionally, the bands at 1600 cm⁻¹ and 1500 cm⁻¹ were assigned to C=C functional groups. The bands at 1140 cm⁻¹ and 2325 cm⁻¹ were attributed to C–O and C–C bonds, respectively.

The crystal structure of the activated carbon was analyzed by XRD (Fig. 2). The diffraction pattern shows two peaks at $2\theta = 25^\circ$ and 42° , corresponding to the (002) and (100) planes, indicative of disordered carbon⁵. The broadness of these peaks suggests that the material possesses an amorphous structure¹⁵.

The surface morphology was examined using SEM, as shown in Fig. 3. The images reveal a highly porous material with abundant surface pores. This porous structure enhances adsorption by providing sufficient accessible sites for the target adsorbate molecules.

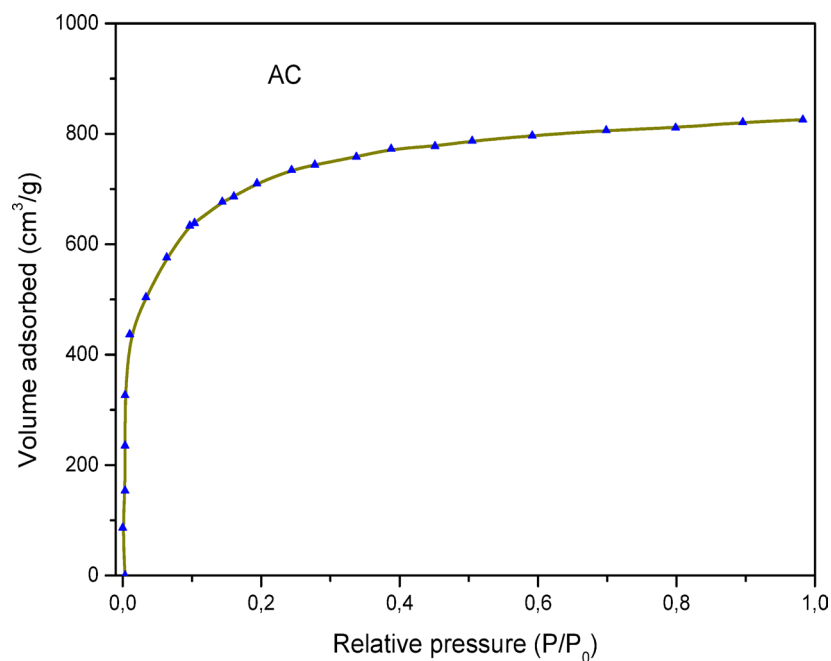


Fig. 4. N₂ isotherm on the AC^{5,26,28}.

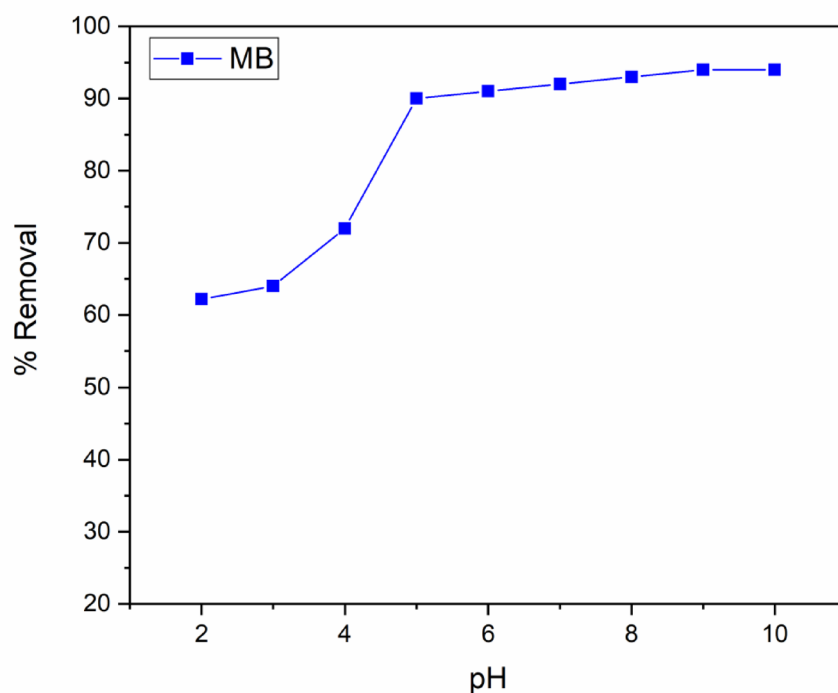


Fig. 5. pH of the MB.

The textural properties of the activated carbon were investigated using N₂ adsorption–desorption isotherms, and the results are summarized in Table 1. According to the IUPAC classification, the isotherm (Fig. 4) corresponds to a type I microporous isotherm⁵. Nitrogen adsorption occurs at low pressures due to the presence of empty sites, and the uptake gradually increases until reaching a plateau at higher pressures, reflecting the occupation of adsorption sites by nitrogen molecules.

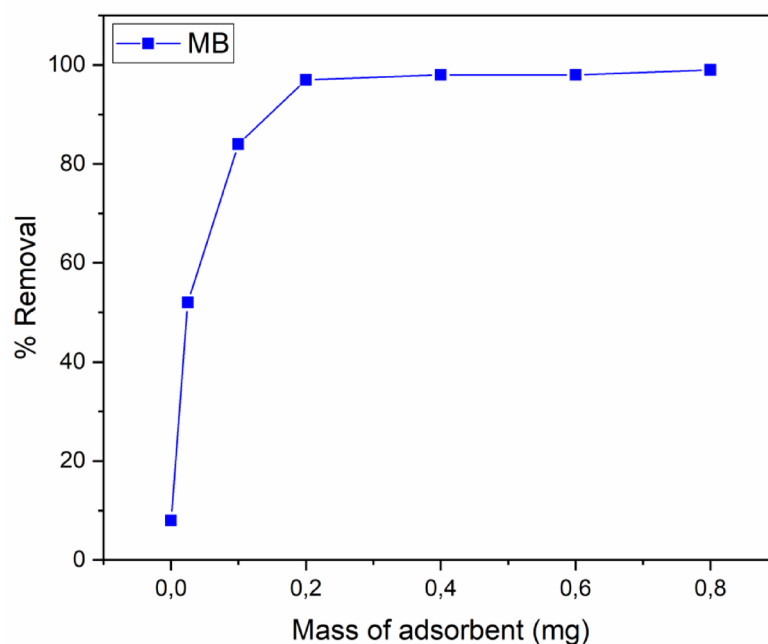


Fig. 6. adsorbent mass on the MB adsorption on the AC.

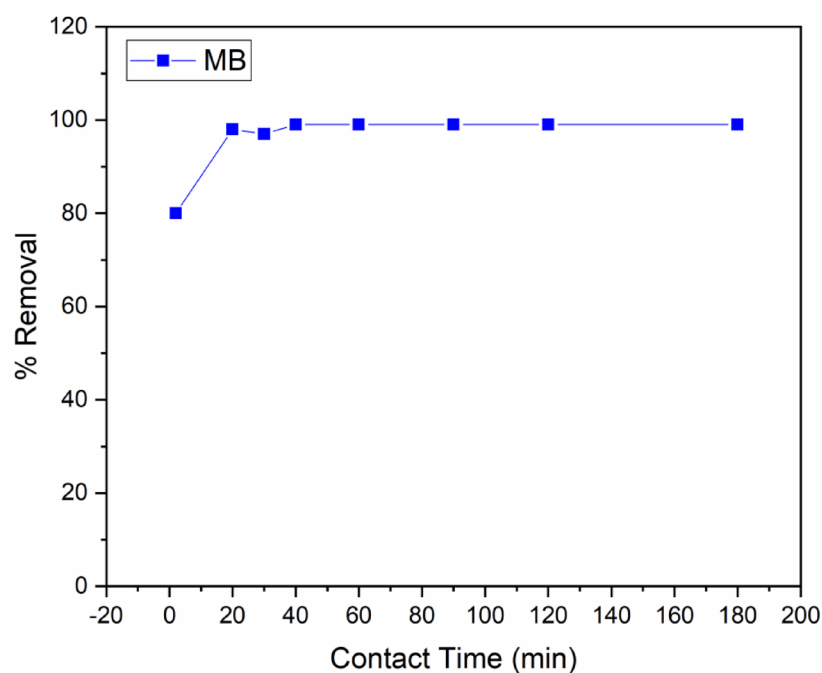


Fig. 7. relationship between MB adsorption and the contact time.

MB adsorption studies on the AC

The pH of a solution critically influences phenol adsorption by affecting the surface charge and interaction between the adsorbent and adsorbate. Under acidic conditions, adsorption is enhanced due to strong electrostatic attraction, whereas at neutral pH, adsorption capacity decreases because of the dominance of positive surface charges¹². Figure 5 presents the removal efficiency of MB on activated carbon at different pH values (2–10). The results indicate that the highest removal efficiency (95%) occurs at pH 2 and decreases significantly to 25% as the pH increases.

The effect of adsorbent mass on MB adsorption was investigated, and the results are shown in Fig. 6. The removal efficiency increased as the adsorbent mass was raised from 0.01 g to 0.1 g, which can be attributed

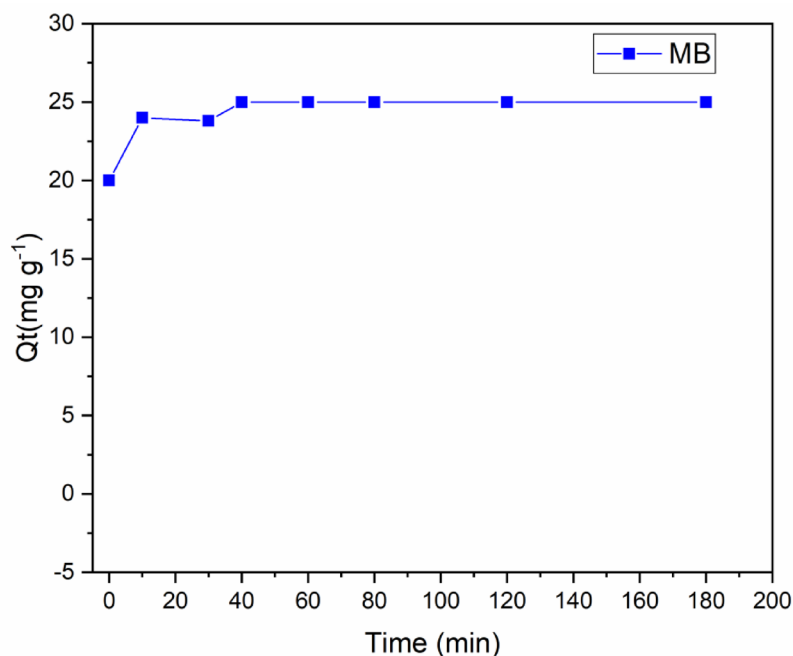


Fig. 8. Adsorption kinetic of MB on the AC.

	MB	
Pseudo 1st order	q_{leexp}	23.21
	k_1 (min^{-1})	0.074
	q_e (mg g^{-1})	0.03
	R^2	0.432
Pseudo 2nd order	k_2 ($\text{g mg}^{-1} \text{min}^{-1}$)	0.091
	Q_e (mg g^{-1})	24.32
	R^2	0.998
	h ($\text{mg g}^{-1} \text{min}^{-1}$)	57.32
	$t_{1/2}$ (min)	0.42

Table 2. Adsorption isotherms parameters.

to the greater specific surface area and availability of adsorption sites⁵. The maximum adsorption efficiency and equilibrium were achieved at an adsorbent mass of 0.2 g, which was therefore used as the optimal mass in subsequent experiments.

The relationship between MB adsorption and contact time is illustrated in Fig. 7. The adsorption efficiency increased with contact time and reached equilibrium after 20 min. During the initial stage, adsorption occurs rapidly due to the abundance of available pores on the activated carbon surface, which facilitates faster mass transfer of MB from the solution to the adsorbent¹⁶.

Adsorption kinetics studies

The kinetic study of MB adsorption on activated carbon, along with the parameters obtained from the pseudo-first- and pseudo-second-order models based on Eqs. (2) and (3), is presented in Fig. 8 and Table 2. The pseudo-second-order model provides the best fit, with a correlation coefficient (R^2) of 0.997. Representative plots of the pseudo-second-order model are shown in Fig. 9. These results indicate that the adsorption of MB on activated carbon is predominantly governed by pseudo-second-order kinetics, suggesting that chemisorption is the main mechanism.

Additionally, the half-adsorption time ($t_{1/2}$) represents the time required to remove 50% of the target compound at equilibrium. A low half-adsorption time reflects a high affinity between the adsorbent and the dye, as evidenced by the rapid removal of most of the dye during the initial adsorption period.

Adsorption isotherms

The Freundlich and Langmuir isotherm models, based on Eqs. (4) and (5), were applied to analyze the equilibrium data for methylene blue (MB) adsorption (Fig. 10). The Langmuir model assumes monolayer adsorption on a

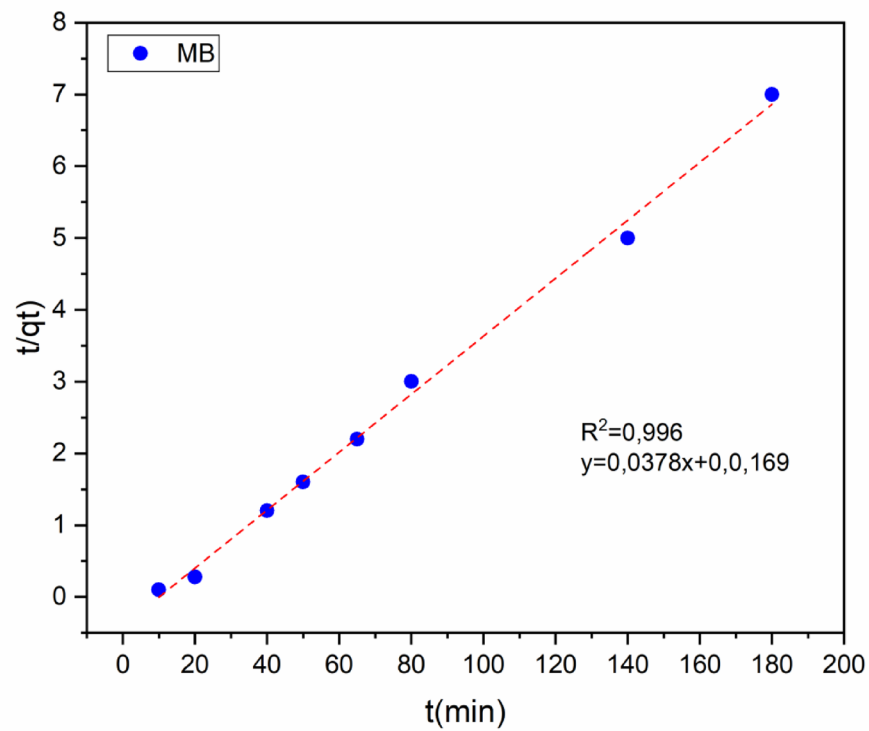


Fig. 9. Pseudo-second order plots for MB.

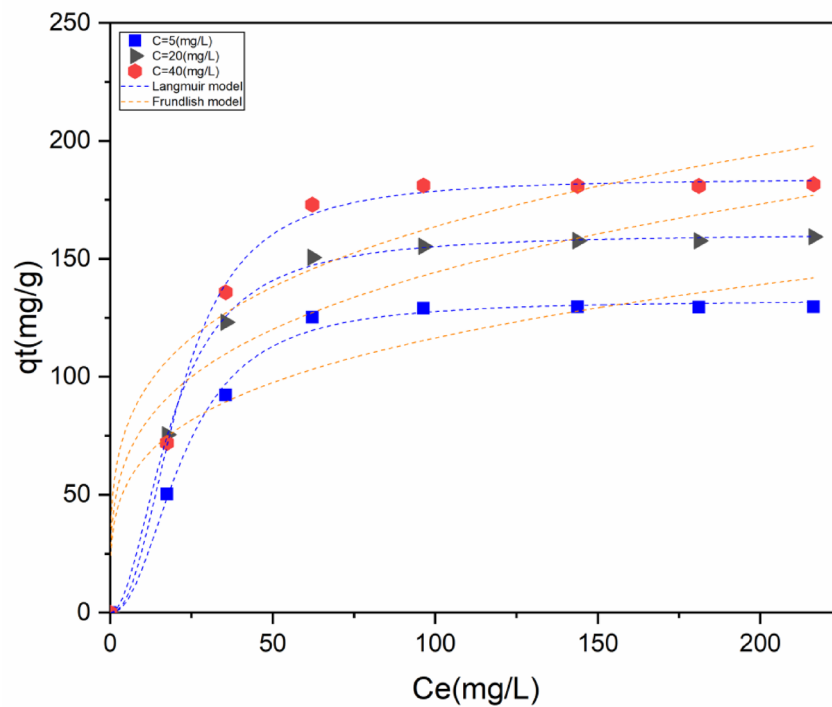


Fig. 10. MB adsorption fitted with Langmuir and Freundlich models.

Isotherms	Parameters	MB
Langmuir	$q_{\max}$ (mg g ⁻¹)	34.3
	K_L (L mg ⁻¹)	1.01
	R_L	0.01
	R^2	0.998
Freundlich	K_F	30.65
	n	6.58
	R^2	0.642

Table 3. Kinetics parameters for pseudo 1st order and pseudo2nd order.

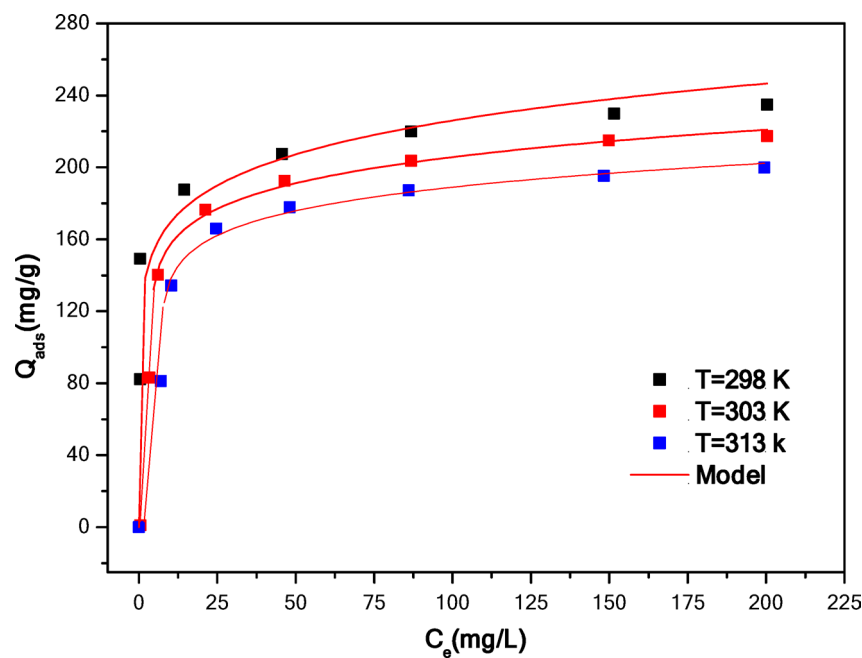


Fig. 11. Adsorption isotherms of MB on AC adsorbent.

Temperature (K)	n	D_m (mg/g)	Q_m (mg/g)	$-\Delta E_1$ (kJ/mol)	$-\Delta E_2$ (kJ/mol)
298	1.78	56.32	235.21	18.90	15.61
303	1.54	60.74	209.62	15.41	11.44
313	1.32	69.17	198.62	11.85	10.42

Table 4. Adjustment parameter values corresponding to the fit model.

homogeneous surface, while the Freundlich model describes multilayer adsorption on a heterogeneous surface¹². The experimental data were evaluated using the R^2 values obtained from the linearized plots of each isotherm model. As shown in Table 3, the Langmuir model provides the best fit, with $R^2=0.993$, indicating that MB adsorption on the activated carbon occurs predominantly as monolayer coverage on a uniform surface.

Physical statistical interpretations

The experimental data and the physical statistical prediction are shown in Fig. 11. The best-fitting advanced statistical physics model for MB adsorption on AC was found to be a multilayer model (Eq. 6). The parameters of this model, presented in Table 4, provide valuable insights into the adsorption mechanism of MB.

- n parameter
This parameter provides information on the orientation of MB molecules on the AC surface. According to the multilayer model^{17,18}, three situations can occur depending on the value of n : when $n < 0.5$, MB molecules interact with at least two adsorption sites, resulting in a parallel tilt of the molecules during adsorption; when

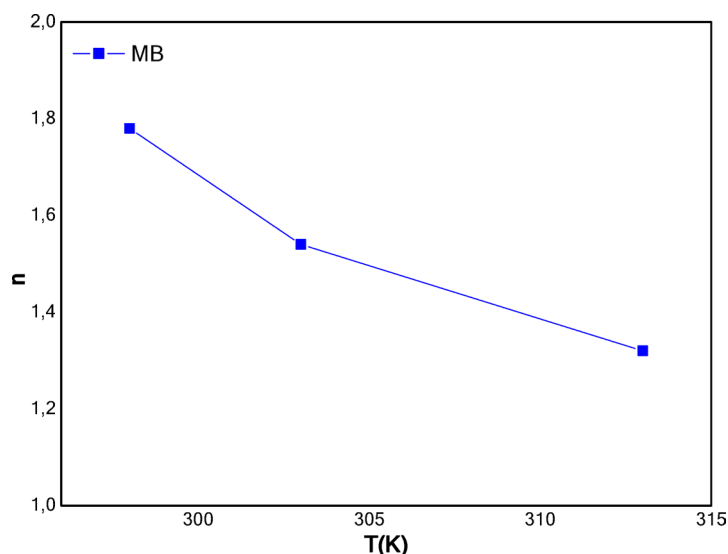


Fig. 12. Effect of temperature on the n parameter for the MB.

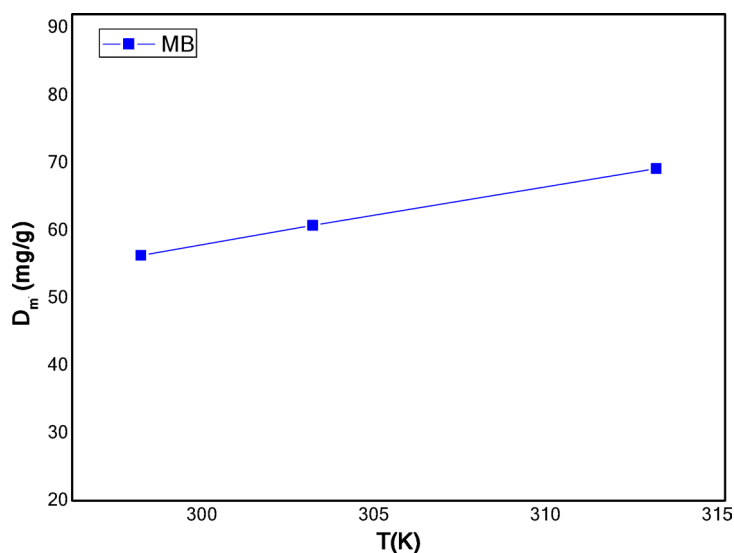


Fig. 13. Effect of temperature on the D_m parameter for the MB.

$n > 1$, the molecules interact with a single adsorption site, indicating a non-parallel orientation; and when $0.5 \leq n \leq 1$, both parallel and non-parallel orientations occur simultaneously in varying proportions.

The calculated n values for MB were greater than 1, indicating that MB molecules predominantly interact with single adsorption sites in a non-parallel orientation. Figure 12 shows the effect of temperature on n ; overall, an increase in temperature reduces n after a slight initial rise at lower temperatures. This decrease in n suggests a shift in MB orientation toward a parallel or mixed inclination on the AC surface¹⁹.

It was observed that the parameter n had values of 1.78, 1.54, and 1.32 at 298, 303, and 313 K, respectively—all greater than unity. This indicates that a vertical molecular orientation and a multi-ionic adsorption mechanism were involved in the interactions between MB ions and the active sites of AC¹⁹. These results suggest that multiple MB ions interacted with a single active site on the adsorbent surface (i.e., a multi-interaction mechanism), leading to a predominantly vertical alignment of the adsorbed dye molecules.

- D_m parameter

This parameter estimates the number of adsorption sites occupied at saturation. Figure 13 illustrates the temperature effect on D_m , which shows an inverse relationship with n . As n decreases with increasing temperature, the adsorption density (D_m) rises, indicating that a lower n corresponds to more occupied adsorption sites¹⁹.

- Q_m parameter

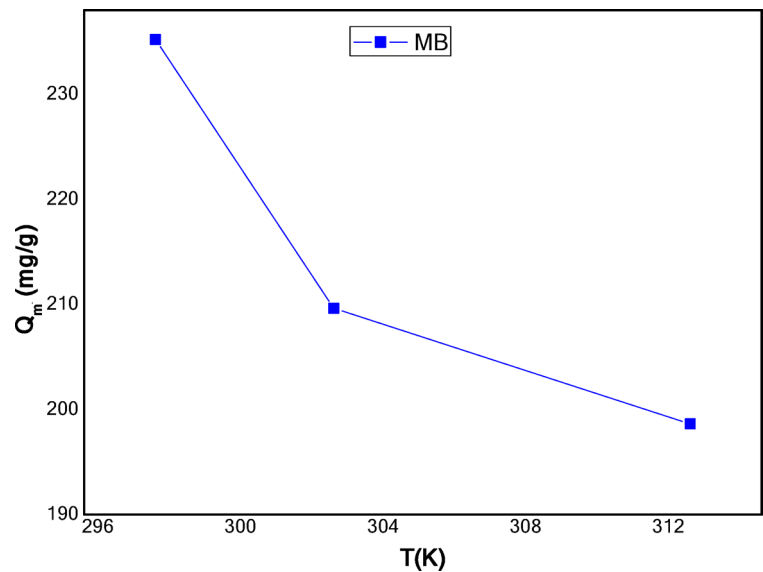


Fig. 14. Effect of temperature on the maximum adsorption capacity.

Adsorbents	Adsorption capacity (mg. g ⁻¹)	References
AC from rice husk waste	27.8	20
AC from black cumin seeds	16.8	21
AC from peanut shell wastes	15.4	22
AC from Parthenium hysterophorus	11.3	23
AC from grape wood	4.8	24
AC from Coriandrum sativum	94.9	7
AC from Calicotome villosa wood	169.8	25
This Work	34.3	

Table 5. Adsorption of various dyes by AC ad sorbents.

The maximum adsorption capacity Q_m depends on the values of n and D_m . Figure 14 shows the temperature effect on Q_m . The decrease of the Q_{sat} values with temperature confirmed the exothermic interaction between MB and AC¹⁹. The Q_{sat} values followed the same trend observed for the parameter n , indicating that the adsorption mechanism of this dye is primarily controlled by n parameter.

- Energetic aspect
The calculation of adsorption energies provides valuable insight into the adsorption mechanisms of MB on AC. Using the half-saturation concentrations and the expressions in Eqs. (7) and (8), the adsorption energies of MB were calculated at different temperatures. The results, summarized in Table 4, show that the negative values of ΔE confirm the exothermic nature of MB adsorption on AC. Moreover, the ΔE values for MB were less than $40\text{ kJ}\cdot\text{mol}^{-1}$, suggesting a physical adsorption process. As expected, the ΔE_1 values were higher than ΔE_2 because ΔE_1 corresponds to interactions between MB and the AC active sites, whereas ΔE_2 reflects MB–MB interactions^{18,19} (Table 5).

Comparison with results reported in the literature

Table 4 presents a comparison between the maximum adsorption capacity ($q_m = 34.4\text{ mg g}^{-1}$) of AC prepared from olive waste and those of other adsorbents reported in the literature. The MB adsorption capacity of AC is higher/lower than that reported for activated carbons prepared from various raw materials. For instance, the highest adsorption capacity reported in the literature was 169.8 mg g^{-1} for activated carbon derived from Calicotome villosa wood²⁵, whereas the lowest value, 4.8 mg g^{-1} , was obtained for activated carbon produced from grape wood²⁴. This wide variation can be attributed primarily to differences in the raw materials used, as well as to variations in pyrolysis conditions and activation methods. Therefore, the adsorption capacity obtained from activated carbon produced using olive waste appears promising, suggesting the need for further studies to optimize its production conditions.

Conclusion

The performance of the activated carbon (AC) adsorbent for MB dye removal was systematically evaluated. The AC was characterized using FTIR, SEM, XRD, and BET techniques. Key factors affecting MB adsorption included pH, adsorbent mass, and contact time. Kinetic studies and adsorption isotherms were analyzed using the pseudo-second-order model and the Langmuir isotherm, respectively. The adsorption mechanism was further investigated using a physical multilayer model, with steric parameters n , D_m , and Q_m evaluated and discussed. Differences in adsorption capacity were found to be primarily influenced by variations in the n parameter. Adsorption energy analysis revealed that MB adsorption was exothermic and occurred via a physisorption process.

Data availability

The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

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Author contributions

Souhail Mohammed bouzgarrou: Conceptualization, data curation, investigation Jedli Hedi, Rym methodology :Validation, visualization. Ehab Sabi: Acquisition Khalifa Slimi: Writing–review & editing.

Competing interests

The authors declare no competing interests.

Additional information

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